INCH-POUND
MIL-DTL-24787/4A(SH)
w/AMENDMENT 1
14 December 2021
SUPERSEDING
MIL-DTL-24787/4A(SH)
24 September 2020

DETAIL SPECIFICATION SHEET

FITTINGS, END, TYPES E AND EL (SPLIT CLAMP), REUSABLE, WITH TYPES SC (SPLIT CLAMP), TF (TAILPIECE FEMALE), AND TM (TAILPIECE MALE) ADAPTERS, FOR FLEXIBLE HOSE ASSEMBLIES

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24787.

1. SCOPE

1.1 Scope. This specification covers reusable split clamp end fittings (types E and EL) and split clamp, female tailpiece, and male tailpiece adapters for hose sizes ranging from -32 to -192 for use with wire- or synthetic fiber-reinforced rubber, flexible hose assemblies in accordance with MIL-DTL-24135 and MIL-DTL-24136.

1.2 Classification. The split clamp end fittings are type E (straight nipple) and EL (90-degree elbow nipple) as identified on figure 1. The adapters are type SC (split clamp assembly), type TF (tailpiece female), and type TM (tailpiece male) as identified on figure 2. For the corresponding hose types, see table I.

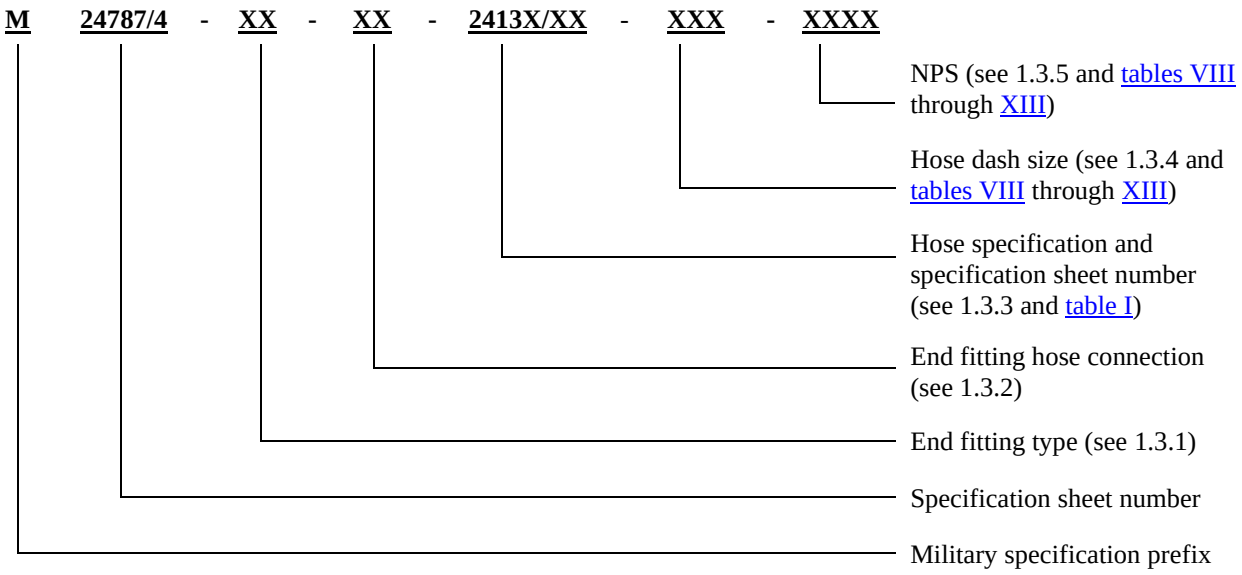
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1.3 Part or identifying number (PIN). The PIN is a definitive part number that is formed by combining the applicable specification sheet number (M24787/4 for MIL-DTL-24787/4), a dash, end fitting type, a dash, end fitting hose connection, hose specification and applicable specification sheet number, a dash, hose dash size, a dash, and nominal pipe size (NPS).



PIN example: M24787/4-EX-SS-24135/04-40X-0250 describes a MIL-DTL-24787/4 end fitting, type EX (split clamp to hose, straight), hose connection "SS" (solid socket), hose type MIL-DTL-24135/4, hose size dash 40 (-40), and an NPS of 2½ inches.

1.3.1 End fitting type. End fitting types are identified in the PIN as follows:

End fitting type	Description	PIN code
E	Split clamp to hose, straight	EX
EL	Split clamp to hose, 90-degree elbow	EL
SC	Split clamp assembly	SC
TF	Tailpiece, female, split clamp, silver braze	TF
TM	Tailpiece, male, split clamp, silver braze	TM

1.3.2 End fitting hose connection. End fitting hose connections are identified in the PIN as follows:

Hose connection	PIN code
Solid Socket	SS
Ring and Band	RB
Bolt Together	BT

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1.3.3 Hose specification and specification sheet number. Specification sheet numbers that only contain one digit will enter "0" as the first digit. For example, for specification sheet 1, the PIN code will be recorded as "01". For end fittings categorized with multiple specification sheets, the first specification and applicable specification sheet will be used for the PIN. For example, for an end fitting that can be used with MIL-DTL-24135/1, MIL-DTL-24135/5, and MIL-DTL-24135/13, the PIN code will be recorded as "24135/01". Where the type of hose is not applicable (for adapters only), all digits will be "X's".

1.3.4 Hose dash size. Hose dash sizes in tables VIII through XIII that only contain two digits will enter "X" as the last digit. For example, for a hose dash size of -40, the PIN code will be recorded as "40X". For hose dash sizes that contain three digits, all digits will be entered in the PIN. For example, for a hose dash size of -128, the PIN code will be recorded as "128". Where the type of hose is not applicable (for adapters only per figure 2, tables XI to XIII), all digits will be "X's".

1.3.5 NPS. When the NPS in tables VIII through XIII is a value that does not contain a fraction, the inch size will be recorded in the first two digits followed by "XX". For inch sizes less than 10, the first digit will contain a zero. For example, for an NPS of 3 inches, the PIN code will be recorded as "03XX". When the NPS is a value of 1 inch or above and contains a fraction, the "X's" below will be replaced with the inch value. For example, for an NPS of 2½ inches, the PIN code will be recorded as "0250". Where NPS is not applicable (for bolt together Type E and EL per table X), all digits will be "X's".

NPS size	PIN code
¼	XX25
½	XX50

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-24135 - Hose, Synthetic Rubber, Wire Reinforced, for Flexible Hose Assemblies, General Specification for

MIL-DTL-24135/4 - Hose, Synthetic Rubber, Wire Reinforced (Double-Wire Braid), for Flexible Hose Assemblies (Medium Pressure, Sizes -40, -48, and -64)

MIL-DTL-24135/10 - Hose, Synthetic Rubber, Wire Reinforced (Single- or Double-Wire Braid), for Flexible Hose Assemblies (Medium Pressure, Sizes -4 through -48)

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MIL-DTL-24135/12	-	Hose, Synthetic Rubber, High Performance, Wire Reinforced (Single-Wire Braid Over Single Synthetic Fiber Braid), for Flexible Hose Assemblies (Medium Pressure, High Temperature, Multi-Fluid, Sizes -4 through -40)
MIL-DTL-24136	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies, General Specification for
MIL-DTL-24136/1	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies (Low Pressure, Sizes -80 through -192)
MIL-DTL-24136/3	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies (Low Pressure, Sizes -4 through -32)
MIL-DTL-24136/4	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies (Low Pressure, Sizes -40, -48, and -64)
MIL-DTL-24787	-	Fittings, End, Reusable, for Flexible Hose Assemblies, General Specification for

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B61	-	Standard Specification for Steam or Valve Bronze Castings
ASTM B148	-	Standard Specification for Aluminum-Bronze Sand Castings
ASTM B271/B271M	-	Standard Specification for Copper-Based Alloy Centrifugal Castings
ASTM B505/B505M	-	Standard Specification for Copper-Based Alloy Continuous Castings
ASTM B584	-	Standard Specification for Copper Alloy Sand Castings for General Applications

(Copies these documents are available online at www.astm.org.)

SAE INTERNATIONAL

DELETED	-	
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SAE AMS7276	-	Rubber: Fluorocarbon (FKM), High-Temperature-Fluid Resistant, Low Compression Set for Seals in Fuel Systems and Specific Engine Oil Systems
SAE AS5857	-	Gland Design, O-Ring and Other Elastomeric Seals, Static Applications
SAE AS5861	-	Retainers, (Backup Rings), Hydraulic and Pneumatic, Polytetrafluoroethylene Resin, Solid, Un-Cut, for Use in Static Glands to AS5857

(Copies of these documents are available online at www.sae.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Split clamp end fittings furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Split clamp end fittings and adapters. Split clamp end fittings and adapters shall be in accordance with the requirements of MIL-DTL-24787 and as specified herein.

3.2.1 Dimensions. End fitting and adapter dimensions shall be in accordance with figures 1 and 2 and tables VIII through XIII (see 6.2). All dimensions in tables VIII through XIII are recorded in inches.

3.2.2 Materials. End fitting and adapter materials shall be in accordance with figures 1 and 2 and tables VIII through XIII.

3.2.3 End fitting availability. Split clamp end fittings (types E and EL) and adapters are available in the hoses and sizes indicated in table I (see 6.2).

TABLE I. Split clamp end fitting by hose specification sheet and size.

End fitting size (hose dash number)	Hose specification sheet			
	24135/4 and /10 (-40 and -48)	24135/10 (-4 through -32), /12, and 24136/3	24136/1	24136/4
-32	--	X	--	--
-40	X	X	--	X
-48	X	--	--	X
-64	X	--	--	X
-80	--	--	X	--
-96	--	--	X	--
-128	--	--	X	--
-160	--	--	X	--
-192	--	--	X	--
NOTE: X denotes where split clamp end fittings are available by hose specification sheet and size.				

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-DTL-24787 and 4.4 of this specification.

4.3 Conformance inspection. Conformance inspection shall be in accordance with MIL-DTL-24787.

4.4 Sampling for qualification testing.

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4.4.1 End fittings. Qualification testing of end fittings shall consist of end fitting samples as specified in table II. For qualification testing, end fittings shall be attached to the appropriate hose indicated in table II.

4.4.2 Use on multiple hoses. Where end fittings are indicated for use on multiple associated hose specification sheets in table I, the basic qualification shall be to the hose associated detail specification having the highest pressure rating. At least one hose sample shall be selected from each additional associated specification sheet for the hydraulic impulse test.

TABLE II. End fitting samples required.

Hose size (dash number)	Number of end fitting samples required 1/	Appropriate hose specification sheet
-40	8	MIL-DTL-24135/4 and /10 (-40 and -48 only)
-64	10	
-8	8	MIL-DTL-24135/10 (-4 to -32), /12, and MIL-DTL-24136/3
-32	10	
-80	8	MIL-DTL-24136/1
-128	10 2/	
-40	6	MIL-DTL-24136/4
-64	8	

NOTES:

1/ When qualifying fittings and hose, the number of end fittings for the small size increases to 10 and large size increases to 14 (see 4.4.4.2 or 4.4.4.3, MIL-DTL-24135 or MIL-DTL-24136, with associated hose detailed specification slant sleet).

2/ If a size larger than -128 is to be offered, then that size shall be represented by 6 sample end fittings.

4.4.3 Hose samples. Hose samples used for the qualification of end fittings shall be qualified to the appropriate associated specification sheet and shall be not less than 15 inches long. The quantity of hose samples required shall be in accordance with table III. Where there is no qualified hose, the prospective end fitting supplier shall work together with a prospective hose supplier to qualify their respective products. (The hose supplier and end fitting supplier may be a single source.)

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TABLE III. Hose samples required.

Hose size (table II)	Number of hose samples to be provided 1/
Smallest	4
Largest	5
2/	3

NOTES:
1/ When qualifying fittings and hose, the number of hoses for -8 increases to 5 and -32 increases to 7 (see 4.4.4.2 or 4.4.4.3 and MIL-DTL-24135 or MIL-DTL-24136, with the appropriate associated hose detailed specification slant sheet).
2/ This sample shall represent the largest hose proposed for qualification (-160 or -192).

4.4.4 Hose and end fitting mix. Where there are two or more qualified suppliers of hose, or when the prospective end fitting supplier is also qualifying an end fitting, the end fitting and hose mix shall be in accordance with 4.4.4.1 through 4.4.4.3 herein, as appropriate. When qualifying a hose and end fitting, hose samples shall be coupled on one end with the end fitting being submitted for qualification and on the other end with a previously qualified end fitting.

4.4.4.1 Qualifying end fittings with multiple qualified hose suppliers. Where there are two or more qualified suppliers of a hose, the hose mix shall be in accordance with table IV.

TABLE IV. Hose mix when there are multiple qualified hose suppliers for end fitting qualification test.

Sample size 1/ (dash number)	Number of hoses to be provided 2/	
	Qualified source #1	Qualified source #2
Smallest	2	2
Largest	3	2
3/	4	2

NOTES:
1/ Satisfactory completion of qualification testing of the samples selected in accordance with tables II, III, and IV will qualify the manufacturer for all sizes in this associated specification sheet for the hose specification sheet(s). Qualification will be extended to all end fittings under the same hose specification sheet(s) under different associated end fitting specification sheets.
2/ See table III for appropriate hose sample.
3/ This sample shall represent the largest hose proposed for qualification (-160 or -192).

4.4.4.2 Qualifying end fittings and hose with one qualified end fitting supplier and one qualified hose supplier. Where the prospective end fitting supplier is also qualifying a hose, and there is only one qualified supplier of end fittings, hose, or both, the end fitting and hose mix shall be in accordance with table V.

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TABLE V. Hose and end fitting mix when there is one qualified supplier for combined qualification test.

Sample size (dash number) 1/	Number of hose samples to be provided 2/		Number of end fittings to be provided	
	Qualifier 3/	Approved 4/ source	Qualifier 3/	Approved 4/ source
Smallest	3	2	6	4
Largest	4	3	8	6

NOTES:

1/ Satisfactory completion of qualification testing of the samples selected in accordance with tables II, III, and V will qualify the manufacturer for all sizes in this associated specification sheet for the hose specification sheet(s). Qualification will be extended to all end fittings under the same hose specification sheet(s) under different associated end fitting specification sheets.

2/ See table III, note 1/, and MIL-DTL-24135 or MIL-DTL-24136, with the appropriate associated hose detailed specification sheet, for appropriate hose sample.

3/ Offeror of hose and end fittings or end fittings to be qualified.

4/ Approved source – previously qualified end fitting and hose, or end fitting or hose.

4.4.4.3 Qualifying end fittings and hose with two or more qualified end fitting and qualified hose suppliers. Where the prospective end fitting supplier is also qualifying a hose and there are two or more qualified suppliers of end fittings, hose, or both, the end fitting and hose mix shall be in accordance with table VI.

TABLE VI. Hose and end fitting mix when there are two or more qualified suppliers for combined qualification test.

Sample size (dash number) 1/	Number of end fittings to be provided		
	Qualifier 2/	Approved 3/ source #1	Approved 4/ source #2
Smallest	6	2	2
Largest	8	3	3
Sample size (dash number) 1/	**Number of hoses to be provided**		
	Qualifier 2/	**Approved 3/ source #1**	**Approved 4/ source #2**
Smallest	3	1	1
Largest	4	2	1

NOTES:

1/ Satisfactory completion of qualification testing of the samples selected in accordance with tables II, III, and VI will qualify the manufacturer for all sizes in this associated specification sheet for the hose specification sheet(s). Qualification will be extended to all end fittings under the same hose specification sheet(s) under different associated end fitting specification sheets.

2/ See table III, note 1/, and MIL-DTL-24135 or MIL-DTL-24136, with the appropriate associated hose detailed specification slant sheet, for appropriate hose sample.

3/ Offeror of hose and end fittings or end fittings to be qualified.

4/ Approved source – previously qualified end fitting and hose, or end fitting or hose.

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4.4.5 Test sequence. The sequence of tests on sample hose assemblies shall be as specified in table VII. If hoses are being qualified at the same time, the sequence of tests shall be in accordance with the applicable associated specification sheet for the hose being qualified. Whichever sequence is used, only those tests in table VII shall apply to split clamp end fittings.

TABLE VII. Sequence of qualification tests on sample assemblies.

Test sequence (see MIL-DTL-24787)	Assembly sample numbers											
	Smallest size				Largest size					Largest size over -128		
	1	2	3	4	5	6	7	8	9	10	11	12
Proof pressure (see 4.5.1)	X	X	X	X	X	X	X	X	X	X	X	X
Oil circulation (see 4.5.2)	--	X	--	--	--	--	X	--	--	X	--	--
Impulse (see 4.5.3) 1/	--	--	--	X	X	--	X	--	--	--	X	--
Burst pressure (see 4.5.4)	X	--	X	--	--	X	--	X	--	--	--	X

NOTE:

1/ Where the end fitting group being qualified is used on multiple hoses (see table II), qualification will be extended to the use of the end fitting on additional hoses provided that at least one sample hose assembly from each identified hose specification sheet is made up and submitted to the hydraulic impulse test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Split clamp end fittings to this specification are intended for use in shipboard or shore-to-ship piping systems where pressures do not exceed the flange rated working pressure and temperatures do not exceed that of the applicable hose.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification sheet.
b. Quantity of end fittings or adapters.
c. Adapter type designation.
d. Hose dash number (if applicable) and/or pipe size (see 3.2.1).
e. End fitting type, class, hose specifications, and size (hose by pipe or tube) (see 3.2.3).
f. Packaging requirements (see 5.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24787 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Subject term (key word) listing.

Elbow

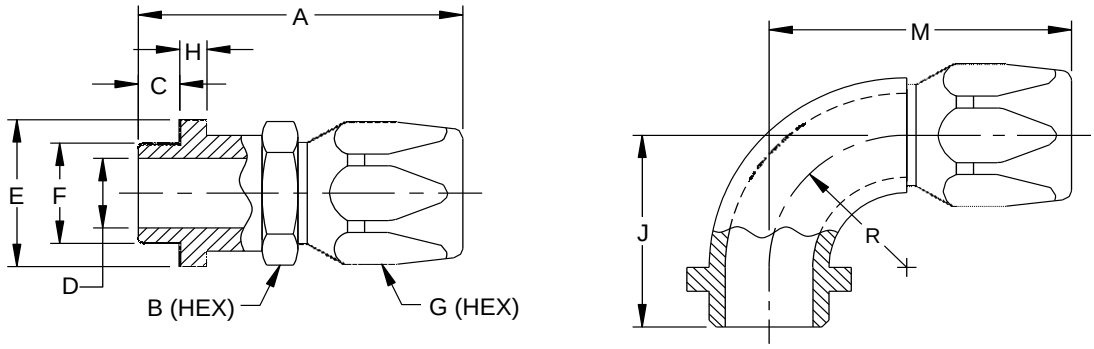
Straight

Synthetic fiber

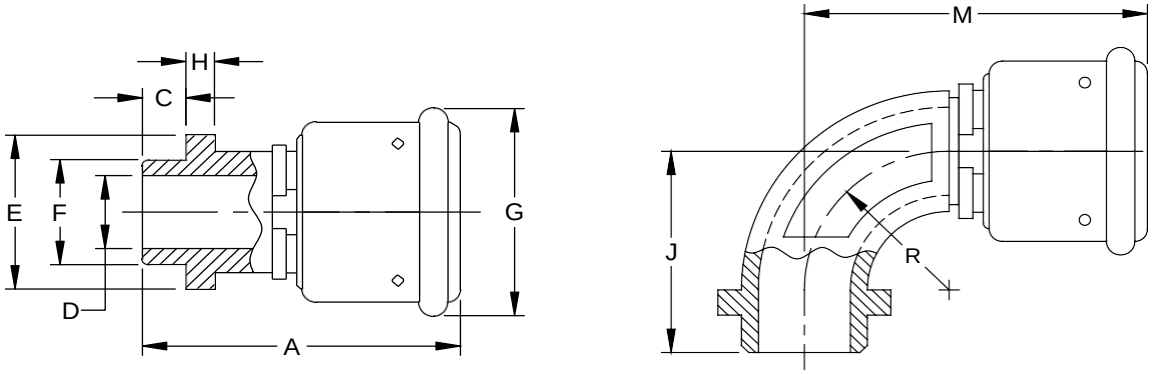
Wire

6.5 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

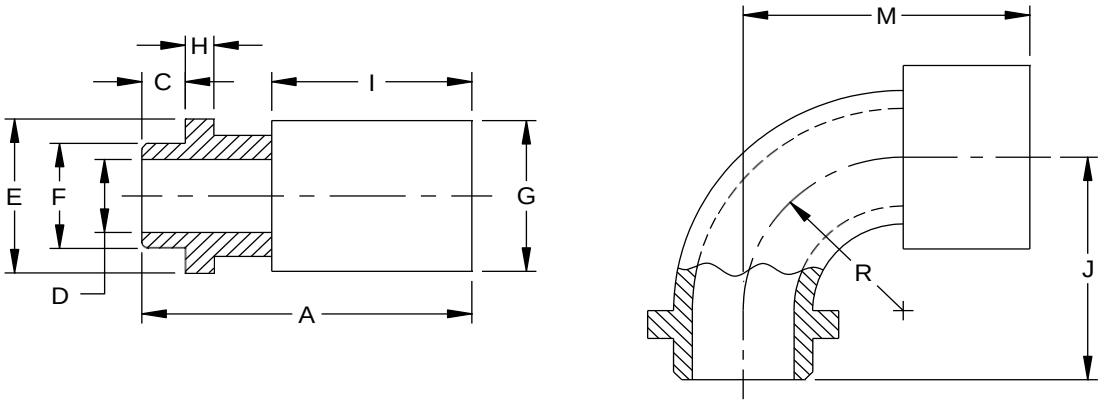
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TYPE E AND EL --------- SOLID SOCKET



TYPE E AND EL --------- RING AND BAND



TYPE E and EL----------BOLT TOGETHER

FIGURE 1. Types E and EL end fittings.

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TABLE VIII. Dimensions for MIL-DTL-24135/10 (-4 to -32), /12, and MIL-DTL-24136/3 solid socket types E and EL.

Hose size (dash number)	NPS	A ref	B ref	C ±0.0025	D ref	E	F ±0.0025	G ref	H ±0.0025	J ref	M ref	R ref
-32	1¼	3.828	2.12	0.3345	1.53	2.433 ±0.005	1.7475	2.516	0.3195	2.81	3.891	1.62
-32	1½	3.796	2.12	0.3345	1.69	2.748 ±0.003	1.9975	2.516	0.3195	2.81	3.891	1.62
-40	2	4.094	2.62	0.3345	2.21	3.248 ±0.003	2.4975	3.266	0.3195	3.04	4.344	1.88
-40	2½	4.188	3.25	0.3345	2.73	3.748 ±0.003	2.9975	3.891	0.3195	3.51	4.563	2.09

NOTES:

1. Material for the nipple shall be UNS C95800 in accordance with ASTM B271/B271M, ASTM B505/B505M, ASTM B584, or ASTM B61.
2. Material for the socket shall be in accordance with QQ-N-281 or QQ-N-286.
3. Dimensions listed as “ref” may vary between qualified manufacturers and have no effect between the fit-up of the nipple and tailpiece.

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TABLE IX. Dimensions for MIL-DTL-24135/4 and /10 (-40 and -48 only) ring and band types E and EL.

Hose size (dash number)	NPS	A ref	C	D ref	E	F ±0.0025	G ref	H ±0.0025	J ref	M ref	R ref
-40	2	4.179	0.3345 ±0.0025	2.05	3.248 ±0.003	2.4975	3.90	0.3195	2.953	5.087	1.88
-48	2½	4.339	0.3345 ±0.0025	2.50	3.748 ±0.003	2.9975	4.50	0.3195	3.031	5.498	2.09
-48	3	4.339	0.3345 ±0.0025	2.50	4.365 ±0.005	3.6225	4.50	0.3195	3.062	5.904	2.09
-64	3½	4.331	0.3345 ±0.0025	3.45	4.930 ±0.005	4.1225	5.44	0.3195	3.906	6.175	2.81
-64	4	4.441	0.404 ±0.005	3.45	5.430 ±0.005	4.6225	5.44	0.3625	4.000	6.175	2.81

NOTES:

1. Material for the nipple shall be UNS C95800 in accordance with ASTM B148, ASTM B271/B271M, or ASTM B505/B505M.
2. Material for the sockets shall be in accordance with QQ-N-281, QQ-N-286, or UNS C95800 in accordance with ASTM B148, ASTM B271/B271M, or ASTM B505/B505M.
3. Dimensions listed as “ref” may vary between qualified manufacturers and have no effect between the fit-up of the nipple and tailpiece.

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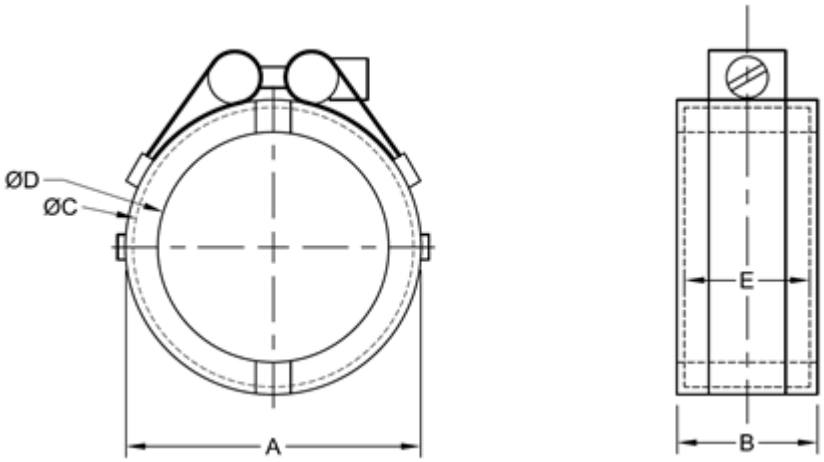
TABLE X. Dimensions for MIL-DTL-24136/1 and /4 bolt together types E and EL.

Hose size (dash number)	NPS	A ref	C	D ref	E	F	G ref 2 segment	G ref 3 segment	H ±0.0025	I ref	J ref	M ref	R ref
-40	2	6.18	0.3345 ±0.0025	2.06	3.248 ±0.003	2.4975 ±0.0025	6.75	5.50	0.3195	3.00	3.50	6.25	2.13
-48	2½	6.18	0.3345 ±0.0025	2.51	3.748 ±0.003	2.9975 ±0.0025	7.75	5.75	0.3195	3.50	3.75	6.88	2.50
-48	3	6.18	0.3345 ±0.0025	2.51	4.365 ±0.005	3.6225 ±0.0025	7.75	5.75	0.3195	3.50	3.75	6.88	2.50
-64	3½	6.31	0.3345 ±0.0025	3.50	4.930 ±0.005	4.1225 ±0.0025	9.00	6.75	0.3195	3.59	4.50	7.00	3.00
-64	4	6.31	0.404 ±0.005	3.50	5.430 ±0.005	4.6225 ±0.0025	9.00	6.75	0.3625	3.59	4.50	7.00	3.00
-80	5	6.22	0.404 ±0.005	4.40	6.630 ±0.005	5.624 ±0.004	11.50	9.25	0.3625	4.75	6.81	8.75	3.50
-96	6	6.58	0.453 ±0.005	5.40	7.567 ±0.005	6.749 ±0.004	12.00	10.50	0.4325	4.75	7.31	9.25	4.00
-128	8	6.80	0.820 ±0.005	7.40	9.970 ±0.005	8.996 ±0.004	14.00	13.25	0.4025	4.75	8.81	10.50	5.00
-160	10	6.84	0.860 ±0.005	9.40	12.057 ±0.005	10.996 ±0.004	16.50	14.75	0.4025	4.75	9.70	12.00	6.38
-192	12	8.16	0.875 ±0.005	11.40	13.375 ±0.005	11.9975 ±0.0025	17.50	17.25	0.4375	6.00	10.70	15.00	7.38

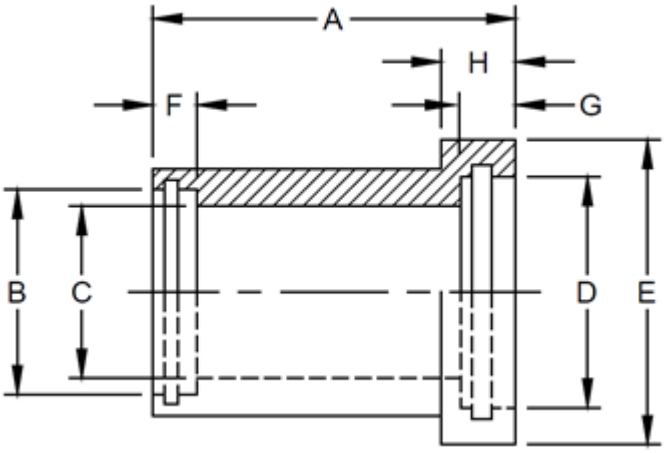
NOTES:

1. Material for the nipple and segments shall be UNS C95800 in accordance with ASTM B148, ASTM B271/B271M, or ASTM B505/B505M.
2. Material for fasteners shall be in accordance with QQ-N-281 or QQ-N-286.
3. Dimensions listed as “ref” may vary between qualified manufacturers and have no effect between the fit-up of the nipple and tailpiece.
4. Bolt together type fittings can be supplied as two-segment bolt together or three-segment bolt together.

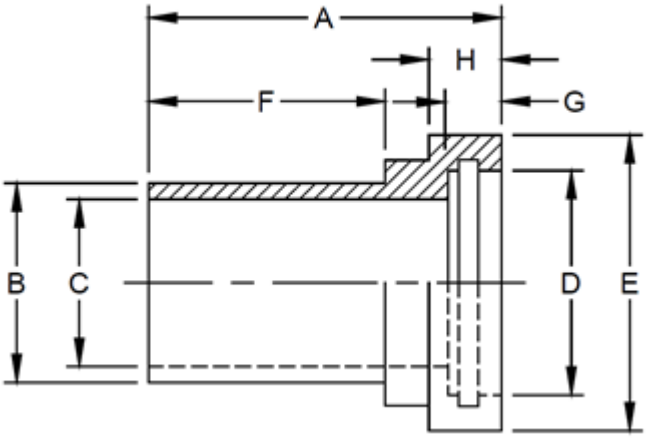
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TYPE SC ------ SPLIT CLAMP ASSEMBLY



TYPE TF -------TAILPIECE, FEMALE



TYPE TM -------TAILPIECE, MALE

FIGURE 2. Split clamp adapters.

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TABLE XI. Dimensions for type SC, all applicable groups.

NPS	A ref	B ref	C	D	E
1¼	3.000	1.310	2.439 +0.002 -0.003	2.125 ±0.010	0.875 ±0.006
1½	3.312	1.310	2.760 ±0.004	2.437 ±0.010	0.875 ±0.006
2	4.00	1.380	3.260 ±0.004	2.937 ±0.010	0.880 ±0.006
2½	4.420	1.440	3.761 ±0.004	3.437 ±0.010	0.937 ±0.006
3	5.188	1.500	4.385 ±0.004	4.062 ±0.010	0.938 ±0.006
3½	5.562	1.500	4.950 ±0.004	4.625 ±0.010	0.938 ±0.006
4	6.125	1.620	5.450 ±0.010	5.110 ±0.016	1.062 ±0.006
5	7.312	1.620	6.650 ±0.010	6.282 ±0.016	1.062 ±0.006
6	8.250	1.940	7.587 ±0.010	7.212 ±0.015	1.312 ±0.006
8	--	--	--	--	--
10	12.879	2.380	12.092 ±0.005	11.632 ±0.015	1.572 ±0.006
12	14.280	2.092	13.415 ±0.0016	12.830 ±0.016	1.350 ±0.016

NOTES:

1. Material for the clamp halves shall be in accordance with ASTM B271, ASTM B505/B505M, ASTM B584, or ASTM B61.
2. Material for the band shall be in accordance with QQ-N-281.
3. Material for the fastener shall be in accordance with QQ-N-281 or QQ-N-286.
4. Dimensions listed as “ref” may vary between qualified manufacturers and have no effect between the fit-up of the nipple and tailpiece.
5. Sizes without dimensional values indicate no qualified product for that size at the time of publication of this document.

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TABLE XII. Type TF, all applicable groups.

NPS	A ref	B +0.007 -0.000	C	D [8/]	E [8/]	F min	G [8/] +0.010 -0.000	H
1¼	1.515	1.660	1.458 ±0.010	1.7575 ±0.0025	2.433 ±0.005	0.500	0.342	0.5435 ±0.0025
1½	1.604	1.900	1.669 ±0.010	2.010 ±0.004	2.748 ±0.003	0.625	0.342	0.5435 ±0.0025
2	1.688	2.375	2.121 ±0.010	2.510 ±0.004	3.248 ±0.003	0.656	0.342	0.5435 ±0.0025
2½	1.890	2.875	2.591 ±0.010	3.010 ±0.004	3.748 ±0.003	0.781	0.342	0.6055 ±0.0025
3	1.547	3.500	3.334 ±0.005	3.638 ±0.005	4.365 ±0.005	0.829	0.342	0.6055 ±0.0025
3½	1.594	4.000	3.810 ±0.005	4.138 ±0.005	4.930 ±0.005	0.875	0.342	0.6055 ±0.0025
4	1.703	4.500	4.310 ±0.005	4.638 ±0.005	5.430 ±0.005	0.907	0.414	0.6875 ±0.0025
5	1.797	5.563	5.295 ±0.005	5.638 ±0.005	6.630 ±0.005	1.001	0.414	0.6875 ±0.0025
6	1.953	6.625	6.375 ±0.005	6.763 ±0.005	7.567 ±0.005	1.109	0.463	0.8675 ±0.0025
8	--	--	--	--	--	--	--	--
10	--	--	--	--	--	--	--	--
12	--	--	--	--	--	--	--	--

NOTES:

1. Material shall be bronze in accordance with ASTM B271/B271M, ASTM B505/B505M, ASTM B584, or ASTM B61or UNS C95800 in accordance with ASTM B148, ASTM B271/B271M, or ASTM B505/B505M.
2. The braze socket and groove shall be designed for ultrasonic inspection.
3. Gland design for the O-ring and back-up ring shall be in accordance with SAE AS5857.
4. The O-ring shall be in accordance with SAE AMS7276.
5. The back-up ring shall be in accordance with SAE AS5861.
6. Dimensions listed as “ref” may vary between qualified manufacturers and have no effect between the fit-up of the nipple and tailpiece.
7. Sizes without dimensional values indicate no qualified product for that size at the time of publication of this document.

[8/] See table XIII. Dimensional values shall be equal in both tables.

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TABLE XIII. Type TM, all applicable groups.

NPS	A ref	B +0.000 -0.007	C	D 7/	E 7/	F +/-0.016	G 7/ +0.010 -0.000	H 7/
1¼	1.963	1.660	1.413 ±0.005	1.7575 ±0.0025	2.433 ±0.005	1.125	0.342	0.5435 ±0.0025
1½	2.088	1.900	1.621 ±0.005	2.010 ±0.004	2.748 ±0.003	1.250	0.342	0.5435 ±0.0025
2	2.183	2.375	2.036 ±0.005	2.510 ±0.004	3.248 ±0.003	1.312	0.342	0.5435 ±0.0025
2½	2.058	2.875	2.586 ±0.005	3.010 ±0.004	3.748 ±0.003	1.125	0.342	0.6055 ±0.0025
3	3.469	3.500	3.000 ±0.016	3.638 ±0.005	4.365 ±0.005	2.505	0.342	0.6055 ±0.0025
3½	3.469	4.000	3.500 ±0.016	4.138 ±0.005	4.930 ±0.005	2.505	0.342	0.6055 ±0.0025
4	3.688	4.500	3.875 ±0.016	4.638 ±0.005	5.430 ±0.005	2.642	0.414	0.6875 ±0.0025
5	3.813	5.563	4.9375 ±0.0155	5.638 ±0.005	6.630 ±0.005	2.767	0.414	0.6875 ±0.0025
6	3.886	6.625	5.9555 ±0.0155	6.763 ±0.005	7.567 ±0.005	2.670	0.463	0.8675 ±0.0025
8	4.386	8.625	7.8755 ±0.0155	9.007 ±0.005	9.970 ±0.005	2.820	0.830	1.150 ±0.005
10	4.566	10.750	9.8155 ±0.0155	11.007 ±0.005	12.057 ±0.005	3.000	0.870	1.150 ±0.005
12	--	--	--	--	--	--	--	--

NOTES:

1. Material shall be bronze in accordance with ASTM B271/B271M, ASTM B505/B505M, ASTM B584, ASTM B61, or UNS C95800 in accordance with ASTM B148, ASTM B271/B271M, or ASTM B505/B505M.
2. Gland design for the O-ring and back-up ring shall be in accordance with SAE AS5857.
3. The O-ring shall be in accordance with SAE AMS7276.
4. The back-up ring shall be in accordance with SAE AS5861.
5. Dimensions listed as “ref” may vary between qualified manufacturers and have no effect between the fit-up of the nipple and tailpiece.
6. Sizes without dimensional values indicate no qualified product for that size at the time of publication of this document.

7/ See table XII. Dimensional values shall be equal in both tables.

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CONCLUDING MATERIAL

Preparing activity:
Navy − SH
(Project 4730-2021-042)

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